Sequences and series of real numbers

1. Sequences of real numbers

Def. 1) $f: \mathbb{N}^* \to \mathbb{R}$ is called seq. of real numbers

Denote: $a_n = f(n)$, $n \geq 1$, $(a_n)_{n \geq 1}$

Def. 2) A seq. a_n is called increasing or decreasing if $a_n \leq a_{n+1}$ $(a_n \geq a_{n+1})$, $\forall n \geq 1$

$a_1 \leq a_2 \leq \dots \leq a_n$ - increasing

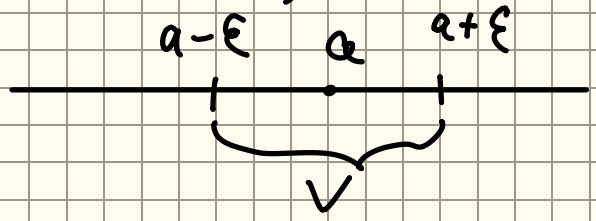
$a_1 \geq a_2 \geq \dots \geq a_n$ - decreasing

Def. 3) a_n is called upper (lower) bounded if $\exists M \in \mathbb{R}$ $(\exists m \in \mathbb{R})$ if $a_n \leq M$ $(a_n \geq m)$, $\forall n \geq 1$. If $m \leq a_n \leq M$, $\forall n \geq 1$, then a_n is called a bounded sequence.

Supremum a_n, $\forall n \geq 1 = M$

Infimum a_n, $\forall n \geq 1 = m$

Def. 4) The $V \subseteq \mathbb{R}$ is called neighbourhood of $a \in \mathbb{R}$ if $\exists (a-\varepsilon, a+\varepsilon) \subseteq V$, $\varepsilon > 0$.

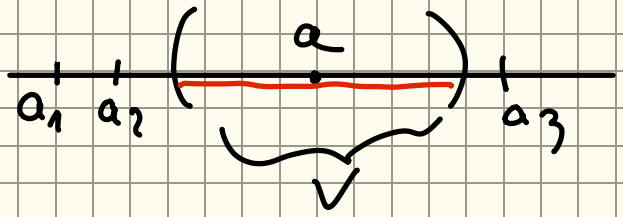


Denote: by $\mathcal{V}(a)$ the set of all neighbourhoods of a.

Denote: $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$

Def. 5) $V \subseteq \mathbb{R}$ is called neigh. of $+\infty$ ($-\infty$) if $\exists$ $(a, +\infty) \subseteq V$ ($\exists$ $(-\infty, a) \subseteq V$)

Def. 6) (Limit) An element $a \in \overline{\mathbb{R}}$ is called limit of (a_n), $n \geq 1$ if $\forall$ $V \in \mathcal{V}(a)$ $\exists$ $n_V \in \mathbb{N}^*$ s.t $a_n \in V$, $\forall$ $n \geq 1$.



Denote: $\lim_{n \to \infty} a_n$ or $\lim a_n$ or $a_n \to a$

Def. 7) A sequence having a finite limit is called convergent.

Th. 8) (a_n), $n \geq 1$ is conv. $\iff$ $\forall \varepsilon > 0, \exists n_\varepsilon \in \mathbb{N}$ s.t. $|a_n - a| < \varepsilon$, $\forall n \geq n_\varepsilon$.

Th. 9) Every monotonic and bounded seq. is convergent (Weirstrass).

Remark: A conv. seq. is bounded but not necessarily monoton.

ex.: $a_n = \frac{(-1)^n}{n} \to 0$

Def. 10) (Subsequence) Let (a_n), $n \geq 1$ and $n_1 < n_2 < n_3 < \dots$ a strictly increasing seq. of natural numbers. Then $a_{n_1}, a_{n_2}, \dots$ denoted by (a_{n_k}), $k \geq 1$ is called subsequence.

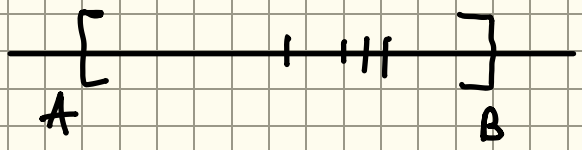
ex.: $a_1, a_3, a_5, \dots$

$a_1, a_4, a_{16}, \dots$

Properties:

- If a seq. has a limit, every subseq. has the same limit.
- Every seq. has a monotonic subseq.

Th. 11) (Cesaro Lemma) Every bounded seq. contains at least one conv. subseq (Bolzano-Weierstrass th.).



Def. 12) (Fundamental seq.) (a_n), $n \geq 1$ is called fundamental seq. if $\forall \varepsilon > 0$, $\exists n_\varepsilon \in \mathbb{N}^*$ s.t. $\forall n \geq n_\varepsilon$, $\forall p \in \mathbb{N}$: $|a_{n+p} - a_n| < \varepsilon$ (Cauchy seq.)

Th. 13) A seq. of real numbers is conv. $\Leftrightarrow$ it is a fundamental seq.

Application: Study the conv. of (a_n), $n \geq 1$

$$a_n = \frac{\sin 1}{1^2} + \frac{\sin 2}{2^2} + \dots + \frac{\sin n}{n^2}, \quad n \geq 1$$

$$|a_{n+p} - a_n| \xrightarrow{n \to \infty} 0, \quad \forall p \in \mathbb{N}$$

$$|a_{n+p} - a_n| = \left| \frac{\sin(n+1)}{(n+1)^2} + \dots + \frac{\sin(n+p)}{(n+p)^2} \right| \leq \left| \frac{\sin(n+1)}{(n+1)^2} \right| + \dots +$$

$$\left|\frac{\sin(n+1)}{(n+1)^2}\right| \le \frac{1}{(n+1)^2} + \dots + \frac{1}{(n+p)^2} \le \frac{1}{n(n+1)} + \frac{1}{(n+1)(n+2)} + \dots + \frac{1}{(n+p-1)(n+p)}$$

$$= \frac{1}{n} - \frac{1}{n+1} + \frac{1}{n+1} - \frac{1}{n+2} + \dots + \frac{1}{n+p-1} - \frac{1}{n+p} = \frac{1}{n} - \frac{1}{n+p} < \frac{1}{n}$$

$|a_{n+p} - a_n| < \frac{1}{n}$, $\forall n, p \in \mathbb{N}^* \Rightarrow |a_{n+p} - a_n| \xrightarrow{n \to \infty} 0$

$\Rightarrow (a_n)$ is fundamental $\Rightarrow (a_n)$ is conv.

2. Upper and Lower limit

Def. 1) $a \in \overline{\mathbb{R}}$ is called limit point (cluster point or accumulation point) of (a_n), $n \ge 1$ if $\exists\, (a_{n_k})$, $k \ge 1$ s.t. $a_{n_k} \xrightarrow[k \to \infty]{} a$

Denote: $L(a_n)$ the set of all limit points of (a_n), $n \ge 1$.

Def. 2) $\limsup a_n = \sup L(a_n)$, $n \in \mathbb{N}^*$

$\liminf a_n = \inf L(a_n)$, $n \in \mathbb{N}^*$

($\overline{\lim}\, a_n$ resp. $\underline{\lim}\, a_n$)

ex.: 1) $a_n = (-1)^{n-1}$, $n \ge 1$

$1, -1, 1, -1, \dots$

$a_{2n} = -1 \to -1$

$a_{2n-1} = 1 \to 1$

$\Rightarrow L(a_n) = \{-1, 1\} \Rightarrow$

$\limsup a_n = 1$

$\liminf a_n = -1$

2) Consider the sequence:

$1, \frac{1}{2}, 1, \frac{1}{2}, \frac{1}{3}, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$

$L(a_n) = \{1, \frac{1}{2}, \frac{1}{3}, \dots\} \cup \{0\}$

$\lim\inf a_n = 0$

$\lim\sup a_n = 1$

3) $(a_n)_{n\geq 1}$, $a_n = \sin n$

$L(a_n) = [-1, 1]$

$\lim\sup a_n = 1$

$\lim\inf a_n = -1$

Th. 3) (a_n), $n \geq 1$ has limit $\iff \lim\sup a_n = \lim\inf a_n$.

3. Series of numbers

(a_n), $n \geq 1$ in a neg. and consider

$\sum_{n=1}^{\infty} a_n = a_1 + a_2 + \dots$

ex.: 1) $\sum_{n=1}^{\infty} \frac{n}{2^n} = \frac{1}{2} + \frac{2}{2^2} + \dots$

2) $\sum_{n=1}^{\infty} (-1)^{n-1} = 1 - 1 + 1 - 1 + \dots$

$(1-1) + (1-1) + \dots = 0 + 0 + 0 + \dots = 0$

$1 + (1-1) + (1-1) + \dots = 1 + 0 + 0 + \dots = 1$

Obs. Properties of finite sums used for infinite sums lead to contradictions.

Def. 1) Let $\sum_{n=1}^{\infty} a_n$ be a series. a_n is called

general term. Consider $(s_n)_{n\geq 1}$:

$s_n = a_1 + a_2 + \dots + a_n$, $n \geq 1$ (seq. of partial sums)

Def. 2) If $\exists \lim s_n = s$, $s \in \overline{\mathbb{R}}$, then s is called sum of the series.

Denote: $\sum_{n=1}^{\infty} a_n = s$

Obs. If s is finite then the series is called convergent.

Ex.: 1) Geometric series

$\sum_{n=0}^{\infty} q^n = 1 + q + q^2 + \dots$, $q \in \mathbb{R}$ is convergent

$\Leftrightarrow |q| < 1 \Rightarrow \sum_{n=0}^{\infty} q^n = \frac{1}{1-q}$

Proof: $s_n = \sum_{n=0}^{\infty} q^n = \begin{cases} \frac{1-q^{n+1}}{1-q}, & q \neq 1 \\ n+1, & q = 1 \end{cases}$

2) Harmonic series

$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \dots = \infty$

3) Generalized harmonic series

$\sum_{n=1}^{\infty} \frac{1}{n^\alpha} = \frac{1}{1^\alpha} + \frac{1}{2^\alpha} + \dots$, $\alpha \in \mathbb{R}$ is convergent

$\Leftrightarrow \alpha > 1$

Denote: $\sum_{n=1}^{\infty} \frac{1}{n^\alpha} = \zeta(\alpha)$, ζ - zeta function, $\alpha > 1$

Particular cases:

$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ (Euler)

$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$